```
✅ Configuration constants loaded successfully
✅ Config module loaded successfully
🤖 Starting Real Arbitrum DeFi Agent
==================================================

Choose operation mode:
1. 🤖 Autonomous mode (bot runs automatically)
2. 🎛️ Manual mode (you control each action)
3. 🌐 Web dashboard (browser interface)
Enter choice (1-3): 1
Loaded configuration: {'learning_rate': 0.01, 'exploration_rate': 0.3400000000000002,
'max_iterations_per_run': 100, 'optimization_target_threshold': 0.95}
🤖 Initializing Arbitrum Testnet Agent...
DEBUG: WALLET_PRIVATE_KEY loaded from environment: [REDACTED]
🔍 DEBUG: Starting RPC manager initialization...
🔍 DEBUG: Network mode: mainnet
DEBUG: ALCHEMY_RPC_URL loaded from environment:
https://arb-mainnet.g.alchemy.com/v2/6ZvYzOV1E80R-bM9XgIIU
🔍 DEBUG: ALCHEMY_RPC_URL from env:
https://arb-mainnet.g.alchemy.com/v2/6ZvYzOV1E80R-bM9XgIIU
🔗 DEBUG: Added Alchemy RPC to endpoints list:
https://arb-mainnet.g.alchemy.com/v2/6ZvYzOV1E80R-...
🔍 DEBUG: Total RPC endpoints to test: 5
  1. https://arb-mainnet.g.alchemy.com/v2/6ZvYzOV1E80R-bM9XgIIU...
  2. https://arbitrum-mainnet.infura.io/v3/5d36f0061cbc4dda980f93...
  3. https://arb1.arbitrum.io/rpc...
  4. https://arbitrum-one.public.blastapi.io...
  5. https://arbitrum-one.publicnode.com...
🔍 Testing RPC: https://arb-mainnet.g.alchemy.com/v2/6ZvYzOV1E80R-bM9XgIIU
✅ RPC passed tests: https://arb-mainnet.g.alchemy.com/v2/6ZvYzOV1E80R-bM9XgIIU (0.25s)
🔍 Testing RPC: https://arbitrum-mainnet.infura.io/v3/5d36f0061cbc4dda980f938ff891c141
✅ RPC passed tests:
https://arbitrum-mainnet.infura.io/v3/5d36f0061cbc4dda980f938ff891c141 (0.53s)
🔍 Testing RPC: https://arb1.arbitrum.io/rpc
✅ RPC passed tests: https://arb1.arbitrum.io/rpc (0.37s)
🔍 Testing RPC: https://arbitrum-one.public.blastapi.io
✅ RPC passed tests: https://arbitrum-one.public.blastapi.io (0.13s)
🔍 Testing RPC: https://arbitrum-one.publicnode.com
✅ RPC passed tests: https://arbitrum-one.publicnode.com (0.96s)
📊 RPC Test Results: 5/5 endpoints working
🌐 Operating on Arbitrum Mainnet
🔍 DEBUG: Final RPC selection results:
  Primary RPC: https://arbitrum-one.public.blastapi.io
  Fallback RPCs: 4 available
🚨 NETWORK_MODE from environment: 'mainnet'
🔗 Primary RPC: https://arbitrum-one.public.blastapi.io
```

🔄 Fallback RPCs: 4 available
🔍 DEBUG: Attempting to create Web3 connection with primary RPC: https://arbitrum-one.public.blastapi.io
✅ Successfully connected to primary RPC: https://arbitrum-one.public.blastapi.io
🔍 DEBUG: Web3 connection verified - Chain ID: 42161, Latest Block: 359933344
🔑 Wallet Address: 0x5B823270e3719CDe8669e5e5326B455EaA8a350b
💰 AGENT INITIALIZED WITH WALLET: 0x5B823270e3719CDe8669e5e5326B455EaA8a350b
✅ Private key format validated and normalized
📋 Mainnet Token addresses verified:
  USDC: 0xAF88D065e8c38FAD0AEff3E253e648A15ceE23DC
  WBTC: 0x2f2a2543B76A4166549F7aaB2e75Bef0aefC5B0f
  WETH: 0x82aF49447D8a07e3bd95BD0d56f35241523fBab1
  DAI: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
  Aave Pool: 0x794a61358D6845594F94dc1DB02A252b5b4814aD
💰 Initialized last_collateral_value_usd to 0.0 (will sync with actual position)
📊 Initialized last_collateral_value_usd to: 0.0
🚀 Initializing Real DeFi Integrations...
🌐 Initializing for Arbitrum Mainnet (Chain ID: 42161)
🪙 ARB Token Address (checksummed): 0x912CE59144191C1204E64559FE8253a0e49E6548
🏦 Aave Integration - Contract Address Verification (Arbitrum Mainnet):
  Pool: 0x794a61358D6845594F94dc1DB02A252b5b4814aD
  Data Provider: 0x69FA688f1Dc47d4B5d8029D5a35FB7a548310654
  WETH: 0x82aF49447D8a07e3bd95BD0d56f35241523fBab1
  WBTC: 0x2f2a2543B76A4166549F7aaB2e75Bef0aefC5B0f
  DAI: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
  USDC: 0xAF88D065e8c38FAD0AEff3E253e648A15ceE23DC
  ARB: 0x912CE59144191C1204E64559FE8253a0e49E6548
✅ Pool address properly checksummed
✅ WETH address properly checksummed
✅ WBTC address properly checksummed
✅ DAI address properly checksummed
✅ USDC address properly checksummed
✅ ARB address properly checksummed
✅ All contract addresses validated successfully for Arbitrum Mainnet
🏦 Aave integration initialized for 0x5B823270e3719CDe8669e5e5326B455EaA8a350b
🔄 Uniswap V3 integration initialized
🪙 ARB Address (Arbitrum Sepolia): 0x912CE59144191C1204E64559FE8253a0e49E6548
📊 Health Monitor Addresses:
  User: 0x5B823270e3719CDe8669e5e5326B455EaA8a350b
  Data Provider: 0x69FA688f1Dc47d4B5d8029D5a35FB7a548310654
  ARB Token: 0x912CE59144191C1204E64559FE8253a0e49E6548
📊 Enhanced Aave Health Monitor initialized for 0x5B823270e3719CDe8669e5e5326B455EaA8a350b
✅ Initialized Real Aave, Uniswap, and Health Monitor Integrations.
⛽ Gas Calculator: Arbitrum Mainnet
⛽ Initialized Gas Calculator.

🏦 Initialized Enhanced Borrow Manager.
🚀 Approving USDC for Aave Pool...
🔍 Contract function validation:
  Token contract: 0xAF88D065e8c38FAD0AEff3E253e648A15ceE23DC
  Approve function found: ✅
  Function object: <Function approve>
🔧 Using reduced approval for zero balance: 1000000000000
🔓 Approving 1000000000000 wei for token
0xAF88D065e8c38FAD0AEff3E253e648A15ceE23DC
🔢 Using pending nonce: 1957 for approval
🔢 Attempt 1: Using fresh pending nonce 1957
🔍 Approve parameters validation:
  Spender: 0x794a61358D6845594F94dc1DB02A252b5b4814aD (type: <class 'str'>)
  Amount: 1000000000000 (type: <class 'int'>)
🔍 COMPREHENSIVE Pre-transaction validation:
  ⚠️ Balance check failed: Could not transact with/call contract function, is contract
deployed correctly and chain synced?
  🔧 Proceeding with transaction anyway - this might be an RPC issue
  ✅ Contract compatibility validated, estimated gas: 26423
✅ Token approval transaction sent to network:
0x58f07e2f449e3e5ae038648b46825bd2f9d49d6a0d9a85403fce20e275074451
⏳ Waiting for transaction receipt for
0x58f07e2f449e3e5ae038648b46825bd2f9d49d6a0d9a85403fce20e275074451 (timeout: 300s)...
✅ Token approval successful:
0x58f07e2f449e3e5ae038648b46825bd2f9d49d6a0d9a85403fce20e275074451
✅ USDC approved for Aave with optimized gas
🚀 Approving WBTC for Aave Pool...
🔍 Contract function validation:
  Token contract: 0x2f2a2543B76A4166549F7aaB2e75Bef0aefC5B0f
  Approve function found: ✅
  Function object: <Function approve>
🔧 Using reduced approval for zero balance: 100000000000000
🔓 Approving 100000000000000 wei for token
0x2f2a2543B76A4166549F7aaB2e75Bef0aefC5B0f
🔢 Using pending nonce: 1958 for approval
🔢 Attempt 1: Using fresh pending nonce 1958
🔍 Approve parameters validation:
  Spender: 0x794a61358D6845594F94dc1DB02A252b5b4814aD (type: <class 'str'>)
  Amount: 100000000000000 (type: <class 'int'>)
🔍 COMPREHENSIVE Pre-transaction validation:
  💰 Current token balance: 0.00000000
  ❌ CRITICAL: Zero token balance detected
  💡 SOLUTION: This is the likely cause of approval failures
  🔧 RECOMMENDATION: Fund wallet with this token first
  ⚠️ Proceeding with minimal approval amount for future funding

🔐 Current allowance: 115792089237316195423570985008687907853269984665640564039457584007913129618847
✅ Sufficient allowance already exists (115792089237316195423570985008687907853269984665640564039457584007913129618847 >= 100000000000)
⚡ Skipping unnecessary approval transaction
✅ WBTC approved for Aave with optimized gas
🚀 Approving WETH for Aave Pool...
🔍 Contract function validation:
Token contract: 0x82aF49447D8a07e3bd95BD0d56f35241523fBab1
Approve function found: ✅
Function object: <Function approve>
🔧 Using reduced approval for zero balance: 1000000000000000000000000
🔓 Approving 1000000000000000000000000 wei for token 0x82aF49447D8a07e3bd95BD0d56f35241523fBab1
🔢 Using pending nonce: 1958 for approval
🔢 Attempt 1: Using fresh pending nonce 1958
🔍 Approve parameters validation:
Spender: 0x794a61358D6845594F94dc1DB02A252b5b4814aD (type: <class 'str'>)
Amount: 1000000000000000000000000 (type: <class 'int'>)
🔍 COMPREHENSIVE Pre-transaction validation:
💰 Current token balance: 0.00000000
❌ CRITICAL: Zero token balance detected
💡 SOLUTION: This is the likely cause of approval failures
🔧 RECOMMENDATION: Fund wallet with this token first
⚠️ Proceeding with minimal approval amount for future funding
🔐 Current allowance: 115792089237316195423570985008687907853269984665640564039457579914298733187402
✅ Sufficient allowance already exists (115792089237316195423570985008687907853269984665640564039457579914298733187402 >= 100000000000000000000)
⚡ Skipping unnecessary approval transaction
✅ WETH approved for Aave with optimized gas
🚀 Approving DAI for Aave Pool...
🔍 Contract function validation:
Token contract: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
Approve function found: ✅
Function object: <Function approve>
🔓 Using infinite approval (MAX_UINT256) - user has balance
🔓 Approving 115792089237316195423570985008687907853269984665640564039457584007913129639935 wei for token 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
🔢 Using pending nonce: 1958 for approval
🔢 Attempt 1: Using fresh pending nonce 1958
🔍 Approve parameters validation:
Spender: 0x794a61358D6845594F94dc1DB02A252b5b4814aD (type: <class 'str'>)

Amount: 115792089237316195423570985008687907853269984665640564039457584007913129639935 (type: <class 'int'>)
🔍 COMPREHENSIVE Pre-transaction validation:
💰 Current token balance: 13.76074703
🔐 Current allowance: 115792089237316195423570985008687907853269984665640564039457584007913129639935
✅ Sufficient allowance already exists (115792089237316195423570985008687907853269984665640564039457584007913129639935 >= 115792089237316195423570985008687907853269984665640564039457584007913129639935)
⚡ Skipping unnecessary approval transaction
✅ DAI approved for Aave with optimized gas
✅ All DeFi integrations initialized successfully!
🚀 Arbitrum agent initialized successfully!
🤝 Collaborative strategy manager ready!

--- Autonomous Agent Loop: Iteration 1 ---

--- Starting New Autonomous Run: 1 ---

🚀 STARTING AUTONOMOUS RUN: 1, ITERATION: 1
==========
📋 Step 1: Network and wallet validation...
🌐 Network Congestion Analysis:
Base Fee: 0.01 gwei
Current Gas: 0.01 gwei
Congestion Ratio: 1.00x
🌐 Network Connected: Block 359933391, Gas Price: 0.01 Gwei
🔗 RPC Endpoint: https://arbitrum-one.public.blastapi.io
⛓️ Chain ID: 42161 (Expected: 42161)
✅ Aave Pool accessible: Revision 8
✅ Network validation passed: Connected and validated
💰 Current ETH balance: 0.003774 ETH
📊 Step 4: Checking Aave account status...
Health Factor: 4.1223
Collateral: $184.48
Available Borrows: $100.52
✅ Conditions favorable for DAI borrowing operations
📊 Predicted success rate: 95.0%
💰 Safe borrow calculation:
Available: $100.52
Safe amount (15%): $10.00
🎯 Attempting safe DAI borrow: $10.00
🏦 Enhanced Borrow Manager: STRICT DAI-ONLY BORROWING
💰 Target amount: $10.00 DAI

🎯 DAI address: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
❌ USDC fallback: DISABLED
✅ Confirmed DAI address: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
💱 USD to DAI conversion: $10.00 = 10000000000000000000 DAI wei
🔍 DAI-specific borrow validation for $10.00
✅ DAI address confirmed: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
🔍 Enhanced validation for $10.00 borrow...
✅ Network connected - Block: 359933393
🔍 Enhanced Validation Results:
  Total Collateral: $184.48
  Total Debt: $35.11
  Available Borrows: $100.52
  Health Factor: 4.1223
  ETH Balance: 0.003774 ETH
✅ All validation checks passed
🔄 DAI borrow attempt 1/3
🎯 Only borrowing DAI: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
🔄 Borrow attempt 1/3
🔧 Getting optimized gas for aave_borrow in normal conditions
⛽ Network gas conditions:
  Current gas price: 0.01 gwei
  Base fee: 0.01 gwei
  Network congestion factor: 1.00
  Total gas multiplier: 1.32
✅ EIP-1559 gas params:
  Gas limit: 600,000
  Max fee: 0.51 gwei
  Priority fee: 0.50 gwei
✅ Borrow successful:
0xdb60235b9afbe8c085079be07e3724a7284cf98e7e654bc61da98d7ea4c0e74c
✅ DAI borrow successful on attempt 1
✅ SUCCESS: Borrowed $10.00 DAI
🔗 Transaction hash: AttributeDict({'blockHash':
HexBytes('0xdcfb431a8974df93850ba49b988c098e4d9cf4b96dfc5e304054f6ae8ba9f9b6'),
'blockNumber': 359933402, 'contractAddress': None, 'cumulativeGasUsed': 643526,
'effectiveGasPrice': 10000000, 'from': '0x5B823270e3719CDe8669e5e5326B455EaA8a350b',
'gasUsed': 362430, 'gasUsedForL1': '0x13fb', 'l1BlockNumber': '0x15e6d26', 'logs':
[AttributeDict({'address': '0x8619d80FB0141ba7F184CbF22fd724116D9f7ffC', 'topics':
[HexBytes('0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'),
HexBytes('0x0000000000000000000000000000000000000000000000000000000000000000'),
HexBytes('0x000000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b')],
'data':
HexBytes('0x0000000000000000000000000000000000000000000000008ac7230489e80000'),
'blockNumber': 359933402, 'transactionHash':
HexBytes('0xdb60235b9afbe8c085079be07e3724a7284cf98e7e654bc61da98d7ea4c0e74c'),
'transactionIndex': 3, 'blockHash':

HexBytes('0xdcfb431a8974df93850ba49b988c098e4d9cf4b96dfc5e304054f6ae8ba9f9b6'),
'logIndex': 0, 'removed': False}), AttributeDict({'address':
'0x8619d80FB0141ba7F184CbF22fd724116D9f7ffC', 'topics':
[HexBytes('0x458f5fa412d0f69b08dd84872b0215675cc67bc1d5b6fd93300a1c3878b86196'),
HexBytes('0x000000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b'),
HexBytes('0x000000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b')],
'data':
HexBytes('0x0000000000000000000000000000000000000000000000008ac7230489e800000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000409733dfee81c57c08b9317'), 'blockNumber': 359933402,
'transactionHash':
HexBytes('0xdb60235b9afbe8c085079be07e3724a7284cf98e7e654bc61da98d7ea4c0e74c'),
'transactionIndex': 3, 'blockHash':
HexBytes('0xdcfb431a8974df93850ba49b988c098e4d9cf4b96dfc5e304054f6ae8ba9f9b6'),
'logIndex': 1, 'removed': False}), AttributeDict({'address':
'0x794a61358D6845594F94dc1DB02A252b5b4814aD', 'topics':
[HexBytes('0x804c9b842b2748a22bb64b345453a3de7ca54a6ca45ce00d415894979e22897a'),
HexBytes('0x000000000000000000000000da10009cbd5d07dd0cecc66161fc93d7c9000da1')],
'data':
HexBytes('0x0000000000000000000000000000000000000000023ba1e715a19a664dd5dc80000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000034ccb3ffa36a907168cd7000000000000000000000000000000000000
000003bf393779b050b33423f8770000000000000000000000000000000000000000409733dfee8
1c57c08b9317'), 'blockNumber': 359933402, 'transactionHash':
HexBytes('0xdb60235b9afbe8c085079be07e3724a7284cf98e7e654bc61da98d7ea4c0e74c'),
'transactionIndex': 3, 'blockHash':
HexBytes('0xdcfb431a8974df93850ba49b988c098e4d9cf4b96dfc5e304054f6ae8ba9f9b6'),
'logIndex': 2, 'removed': False}), AttributeDict({'address':
'0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1', 'topics':
[HexBytes('0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'),
HexBytes('0x000000000000000000000000082e64f49ed5ec1bc6e43dad4fc8af9bb3a2312ee'),
HexBytes('0x000000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b')],
'data':
HexBytes('0x0000000000000000000000000000000000000000000000008ac7230489e80000'),
'blockNumber': 359933402, 'transactionHash':
HexBytes('0xdb60235b9afbe8c085079be07e3724a7284cf98e7e654bc61da98d7ea4c0e74c'),
'transactionIndex': 3, 'blockHash':
HexBytes('0xdcfb431a8974df93850ba49b988c098e4d9cf4b96dfc5e304054f6ae8ba9f9b6'),
'logIndex': 3, 'removed': False}), AttributeDict({'address':
'0x794a61358D6845594F94dc1DB02A252b5b4814aD', 'topics':
[HexBytes('0xb3d084820fb1a9decffb176436bd02558d15fac9b0ddfed8c465bc7359d7dce0'),
HexBytes('0x000000000000000000000000da10009cbd5d07dd0cecc66161fc93d7c9000da1'),
HexBytes('0x000000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b'),
HexBytes('0x0000000000000000000000000000000000000000000000000000000000000000')],
'data':
HexBytes('0x000000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b0000

0000000000000000000000000000000000000000008ac7230489e800000000000000000000000000
0000000000000000000000000000000000000000000200000000000000000000000000000000000000
00000034ccb3ffa36a907168cd70'), 'blockNumber': 359933402, 'transactionHash':
HexBytes('0xdb60235b9afbe8c085079be07e3724a7284cf98e7e654bc61da98d7ea4c0e74c'),
'transactionIndex': 3, 'blockHash':
HexBytes('0xdcfb431a8974df93850ba49b988c098e4d9cf4b96dfc5e304054f6ae8ba9f9b6'),
'logIndex': 4, 'removed': False})], 'logsBloom':
HexBytes('0x0008000000004000000000000000000000000000000000000000000000000000000000
0000000000000000000000400010000000400004000000000000000000800000000000000100
0008000000000000400000000000000000400000000000000000200000000000000000008000000
0000000000000000201000000000000000000000000000000010000000000000000000800000000000
0000000000000000000000000000000000000000010000100000000000000800000000000001000200
0002000001000000002000000008200000000000040000000000000200000000000200000000008
0000000100000000000000000000000000000000000'), 'status': 1, 'timeboosted': False, 'to':
'0x794a61358D6845594F94dc1DB02A252b5b4814aD', 'transactionHash':
HexBytes('0xdb60235b9afbe8c085079be07e3724a7284cf98e7e654bc61da98d7ea4c0e74c'),
'transactionIndex': 3, 'type': 0})
✅ DAI borrow successful!
✅ Operation 'borrow' recorded. Success rate: 100.0%
Next operation available in 60s

📊 AUTONOMOUS TASK SUMMARY:
Operations completed: 5
Performance score: 0.600
Run: 1, Iteration: 1
Logged performance: Run 1, Iteration 1, Metric: 0.6000000000000001

🤝 COLLABORATIVE STRATEGY ANALYSIS:
⚠️ Agent missing get_recent_performance method: 'ArbitrumTestnetAgent' object has no attribute 'get_recent_performance'
📭 No pending proposals
✅ Iteration 1 completed successfully. Waiting for next cycle.
💤 Sleeping for 45 seconds before next iteration...

--- Autonomous Agent Loop: Iteration 2 ---

🚀 STARTING AUTONOMOUS RUN: 1, ITERATION: 2
==========
📋 Step 1: Network and wallet validation...
🌐 Network Congestion Analysis:
Base Fee: 0.01 gwei
Current Gas: 0.01 gwei
Congestion Ratio: 1.00x
🌐 Network Connected: Block 359933582, Gas Price: 0.01 Gwei
🔗 RPC Endpoint: https://arbitrum-one.public.blastapi.io
⛓️ Chain ID: 42161 (Expected: 42161)

✅ Aave Pool accessible: Revision 8
✅ Network validation passed: Connected and validated
💰 Current ETH balance: 0.003769 ETH
📊 Step 4: Checking Aave account status...
  Health Factor: 3.6139
  Collateral: $208.24
  Available Borrows: $105.49
✅ Conditions favorable for DAI borrowing operations
📊 Predicted success rate: 95.0%
⏰ Operation in cooldown. 15s remaining
⏰ Operations in cooldown period

📊 AUTONOMOUS TASK SUMMARY:
  Operations completed: 3
  Performance score: 0.300
  Run: 1, Iteration: 2
Logged performance: Run 1, Iteration 2, Metric: 0.30000000000000004

🤝 COLLABORATIVE STRATEGY ANALYSIS:
⚠️ Agent missing get_recent_performance method: 'ArbitrumTestnetAgent' object has no attribute 'get_recent_performance'
📭 No pending proposals
✅ Iteration 2 completed successfully. Waiting for next cycle.
💤 Sleeping for 45 seconds before next iteration...

--- Autonomous Agent Loop: Iteration 3 ---

🚀 STARTING AUTONOMOUS RUN: 1, ITERATION: 3
==========
📋 Step 1: Network and wallet validation...
🌐 Network Congestion Analysis:
  Base Fee: 0.01 gwei
  Current Gas: 0.01 gwei
  Congestion Ratio: 1.00x
🌐 Network Connected: Block 359933763, Gas Price: 0.01 Gwei
🔗 RPC Endpoint: https://arbitrum-one.public.blastapi.io
⛓️ Chain ID: 42161 (Expected: 42161)
✅ Aave Pool accessible: Revision 8
✅ Network validation passed: Connected and validated
💰 Current ETH balance: 0.003769 ETH
📊 Step 4: Checking Aave account status...
  Health Factor: 3.6139
  Collateral: $208.24
  Available Borrows: $105.49
✅ Conditions favorable for DAI borrowing operations
📊 Predicted success rate: 95.0%

💰 Safe borrow calculation:
Available: $105.49
Safe amount (15%): $10.00
🎯 Attempting safe DAI borrow: $10.00
🏦 Enhanced Borrow Manager: STRICT DAI-ONLY BORROWING
💰 Target amount: $10.00 DAI
🎯 DAI address: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
❌ USDC fallback: DISABLED
✅ Confirmed DAI address: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
💱 USD to DAI conversion: $10.00 = 10000000000000000000 DAI wei
🔍 DAI-specific borrow validation for $10.00
✅ DAI address confirmed: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
🔍 Enhanced validation for $10.00 borrow...
✅ Network connected - Block: 359933765
🔍 Enhanced Validation Results:
Total Collateral: $208.24
Total Debt: $45.11
Available Borrows: $105.49
Health Factor: 3.6139
ETH Balance: 0.003769 ETH
✅ All validation checks passed
🔄 DAI borrow attempt 1/3
🎯 Only borrowing DAI: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
🔄 Borrow attempt 1/3
🔧 Getting optimized gas for aave_borrow in normal conditions
⛽ Network gas conditions:
Current gas price: 0.01 gwei
Base fee: 0.01 gwei
Network congestion factor: 1.00
Total gas multiplier: 1.32
✅ EIP-1559 gas params:
Gas limit: 600,000
Max fee: 0.51 gwei
Priority fee: 0.50 gwei
✅ Borrow successful:
0xbb8502e4e60dc4b53cdea3527fd49b75b58e6dc785a2bbd224c38904f38cd5ad
✅ DAI borrow successful on attempt 1
✅ SUCCESS: Borrowed $10.00 DAI
🔗 Transaction hash: AttributeDict({'blockHash':
HexBytes('0xf006a18cd700040c4d705cd55fb33016187398ca55c0933fb990a7e5a7f9fcd0'),
'blockNumber': 359933774, 'contractAddress': None, 'cumulativeGasUsed': 565833,
'effectiveGasPrice': 10000000, 'from': '0x5B823270e3719CDe8669e5e5326B455EaA8a350b',
'gasUsed': 389784, 'gasUsedForL1': '0x1422', 'l1BlockNumber': '0x15e6d2e', 'logs':
[AttributeDict({'address': '0x8619d80FB0141ba7F184CbF22fd724116D9f7ffC', 'topics':
[HexBytes('0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'),
HexBytes('0x0000000000000000000000000000000000000000000000000000000000000000'),

HexBytes('0x000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b')],
'data':
HexBytes('0x0000000000000000000000000000000000000000000008ac724baba5a027a'),
'blockNumber': 359933774, 'transactionHash':
HexBytes('0xbb8502e4e60dc4b53cdea3527fd49b75b58e6dc785a2bbd224c38904f38cd5ad'),
'transactionIndex': 3, 'blockHash':
HexBytes('0xf006a18cd700040c4d705cd55fb33016187398ca55c0933fb990a7e5a7f9fcd0'),
'logIndex': 4, 'removed': False}), AttributeDict({'address':
'0x8619d80FB0141ba7F184CbF22fd724116D9f7ffC', 'topics':
[HexBytes('0x458f5fa412d0f69b08dd84872b0215675cc67bc1d5b6fd93300a1c3878b86196'),
HexBytes('0x000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b'),
HexBytes('0x000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b')],
'data':
HexBytes('0x0000000000000000000000000000000000000000000008ac724baba5a027a0000
000000000000000000000000000000000000000000000001b63072027a0000000000000000000000
000000000000000000000409734abe02e7ecb4974a05'), 'blockNumber': 359933774,
'transactionHash':
HexBytes('0xbb8502e4e60dc4b53cdea3527fd49b75b58e6dc785a2bbd224c38904f38cd5ad'),
'transactionIndex': 3, 'blockHash':
HexBytes('0xf006a18cd700040c4d705cd55fb33016187398ca55c0933fb990a7e5a7f9fcd0'),
'logIndex': 5, 'removed': False}), AttributeDict({'address':
'0x794a61358D6845594F94dc1DB02A252b5b4814aD', 'topics':
[HexBytes('0x804c9b842b2748a22bb64b345453a3de7ca54a6ca45ce00d415894979e22897a'),
HexBytes('0x000000000000000000000000da10009cbd5d07dd0cecc66161fc93d7c9000da1')],
'data':
HexBytes('0x0000000000000000000000000000000000000000023b83134db95beb7e7e0630000
000000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000034c9e7bac568e951c8ac7000000000000000000000000000000000000000
000003bf393f7b159a6af4929eaf000000000000000000000000000000000000000409734abe02e
7ecb4974a05'), 'blockNumber': 359933774, 'transactionHash':
HexBytes('0xbb8502e4e60dc4b53cdea3527fd49b75b58e6dc785a2bbd224c38904f38cd5ad'),
'transactionIndex': 3, 'blockHash':
HexBytes('0xf006a18cd700040c4d705cd55fb33016187398ca55c0933fb990a7e5a7f9fcd0'),
'logIndex': 6, 'removed': False}), AttributeDict({'address':
'0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1', 'topics':
[HexBytes('0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'),
HexBytes('0x000000000000000000000000082e64f49ed5ec1bc6e43dad4fc8af9bb3a2312ee'),
HexBytes('0x000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b')],
'data':
HexBytes('0x0000000000000000000000000000000000000000000008ac7230489e80000'),
'blockNumber': 359933774, 'transactionHash':
HexBytes('0xbb8502e4e60dc4b53cdea3527fd49b75b58e6dc785a2bbd224c38904f38cd5ad'),
'transactionIndex': 3, 'blockHash':
HexBytes('0xf006a18cd700040c4d705cd55fb33016187398ca55c0933fb990a7e5a7f9fcd0'),
'logIndex': 7, 'removed': False}), AttributeDict({'address':
'0x794a61358D6845594F94dc1DB02A252b5b4814aD', 'topics':

[HexBytes('0xb3d084820fb1a9decffb176436bd02558d15fac9b0ddfed8c465bc7359d7dce0'),
HexBytes('0x000000000000000000000000da10009cbd5d07dd0cecc66161fc93d7c9000da1'),
HexBytes('0x000000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b'),
HexBytes('0x0000000000000000000000000000000000000000000000000000000000000000')],
'data':
HexBytes('0x000000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b0000
0000000000000000000000000000000000000000008ac7230489e800000000000000000000000000
0000000000000000000000000000000000000000000020000000000000000000000000000000000000
00000034c9e7bac568e951c8ac70'), 'blockNumber': 359933774, 'transactionHash':
HexBytes('0xbb8502e4e60dc4b53cdea3527fd49b75b58e6dc785a2bbd224c38904f38cd5ad'),
'transactionIndex': 3, 'blockHash':
HexBytes('0xf006a18cd700040c4d705cd55fb33016187398ca55c0933fb990a7e5a7f9fcd0'),
'logIndex': 8, 'removed': False})], 'logsBloom':
HexBytes('0x0008000000004000000000000000000000000000000000000000000000000000000000
0000000000000000000000004000100000000400004000000000000000000800000000000000100
0008000000000000040000000000000000040000000000000000020000000000000000000800000
0000000000000000201000000000000000000000000000000100000000000000000000800000000000
0000000000000000000000000000000000000000100001000000000000000800000000000100020
0002000001000000002000000008200000000000040000000000000020000000000020000000008
0000000100000000000000000000000000000000000'), 'status': 1, 'timeboosted': False, 'to':
'0x794a61358D6845594F94dc1DB02A252b5b4814aD', 'transactionHash':
HexBytes('0xbb8502e4e60dc4b53cdea3527fd49b75b58e6dc785a2bbd224c38904f38cd5ad'),
'transactionIndex': 3, 'type': 0})
✅ DAI borrow successful!
✅ Operation 'borrow' recorded. Success rate: 100.0%
Next operation available in 60s

📊 AUTONOMOUS TASK SUMMARY:
Operations completed: 5
Performance score: 0.600
Run: 1, Iteration: 3
Logged performance: Run 1, Iteration 3, Metric: 0.6000000000000001

🤝 COLLABORATIVE STRATEGY ANALYSIS:
⚠️ Agent missing get_recent_performance method: 'ArbitrumTestnetAgent' object has no attribute 'get_recent_performance'
📭 No pending proposals
✅ Iteration 3 completed successfully. Waiting for next cycle.
💤 Sleeping for 45 seconds before next iteration...

--- Autonomous Agent Loop: Iteration 4 ---

🚀 STARTING AUTONOMOUS RUN: 1, ITERATION: 4
==========
📋 Step 1: Network and wallet validation...
🌐 Network Congestion Analysis:

Base Fee: 0.01 gwei
Current Gas: 0.01 gwei
Congestion Ratio: 1.00x
🌐 Network Connected: Block 359933954, Gas Price: 0.01 Gwei
🔗 RPC Endpoint: https://arbitrum-one.public.blastapi.io
⛓️ Chain ID: 42161 (Expected: 42161)
✅ Aave Pool accessible: Revision 8
✅ Network validation passed: Connected and validated
💰 Current ETH balance: 0.003765 ETH
📊 Step 4: Checking Aave account status...
Health Factor: 2.9581
Collateral: $208.24
Available Borrows: $95.49
✅ Conditions favorable for DAI borrowing operations
📊 Predicted success rate: 90.0%
⏰ Operation in cooldown. 15s remaining
⏰ Operations in cooldown period

📊 AUTONOMOUS TASK SUMMARY:
Operations completed: 3
Performance score: 0.300
Run: 1, Iteration: 4
Logged performance: Run 1, Iteration 4, Metric: 0.30000000000000004

🤝 COLLABORATIVE STRATEGY ANALYSIS:
⚠️ Agent missing get_recent_performance method: 'ArbitrumTestnetAgent' object has no attribute 'get_recent_performance'
📭 No pending proposals
✅ Iteration 4 completed successfully. Waiting for next cycle.
💤 Sleeping for 45 seconds before next iteration...

--- Autonomous Agent Loop: Iteration 5 ---

🚀 STARTING AUTONOMOUS RUN: 1, ITERATION: 5
==========
📋 Step 1: Network and wallet validation...
🌐 Network Congestion Analysis:
Base Fee: 0.01 gwei
Current Gas: 0.01 gwei
Congestion Ratio: 1.00x
🌐 Network Connected: Block 359934135, Gas Price: 0.01 Gwei
🔗 RPC Endpoint: https://arbitrum-one.public.blastapi.io
⛓️ Chain ID: 42161 (Expected: 42161)
✅ Aave Pool accessible: Revision 8
✅ Network validation passed: Connected and validated
💰 Current ETH balance: 0.003763 ETH

📊 Step 4: Checking Aave account status...
Health Factor: 3.0977
Collateral: $218.24
Available Borrows: $101.81
✅ Conditions favorable for DAI borrowing operations
📊 Predicted success rate: 95.0%
💰 Safe borrow calculation:
Available: $101.81
Safe amount (15%): $10.00
🎯 Attempting safe DAI borrow: $10.00
🏦 Enhanced Borrow Manager: STRICT DAI-ONLY BORROWING
💰 Target amount: $10.00 DAI
🎯 DAI address: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
❌ USDC fallback: DISABLED
✅ Confirmed DAI address: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
💱 USD to DAI conversion: $10.00 = 10000000000000000000 DAI wei
🔍 DAI-specific borrow validation for $10.00
✅ DAI address confirmed: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
🔍 Enhanced validation for $10.00 borrow...
✅ Network connected - Block: 359934136
🔍 Enhanced Validation Results:
Total Collateral: $218.24
Total Debt: $55.11
Available Borrows: $101.81
Health Factor: 3.0977
ETH Balance: 0.003763 ETH
✅ All validation checks passed
🔄 DAI borrow attempt 1/3
🎯 Only borrowing DAI: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
🔄 Borrow attempt 1/3
🔧 Getting optimized gas for aave_borrow in normal conditions
⛽ Network gas conditions:
Current gas price: 0.01 gwei
Base fee: 0.01 gwei
Network congestion factor: 1.00
Total gas multiplier: 1.32
✅ EIP-1559 gas params:
Gas limit: 600,000
Max fee: 0.51 gwei
Priority fee: 0.50 gwei
✅ Borrow successful:
0x9d3042698153ed0bf339c39533c314e7b1ec1946fd20b0f8324d3b956e8834db
✅ DAI borrow successful on attempt 1
✅ SUCCESS: Borrowed $10.00 DAI
🔗 Transaction hash: AttributeDict({'blockHash':
HexBytes('0x60610ff99eaa20ed9c270f166692331f88371d6e7f51e2043ec92725a163406f'),

'blockNumber': 359934146, 'contractAddress': None, 'cumulativeGasUsed': 485820,
'effectiveGasPrice': 10000000, 'from': '0x5B823270e3719CDe8669e5e5326B455EaA8a350b',
'gasUsed': 389841, 'gasUsedForL1': '0x145b', 'l1BlockNumber': '0x15e6d35', 'logs':
[AttributeDict({'address': '0x8619d80FB0141ba7F184CbF22fd724116D9f7ffC', 'topics':
[HexBytes('0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'),
HexBytes('0x0000000000000000000000000000000000000000000000000000000000000000'),
HexBytes('0x000000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b')],
'data':
HexBytes('0x000000000000000000000000000000000000000000000008ac72670e05a2e80'),
'blockNumber': 359934146, 'transactionHash':
HexBytes('0x9d3042698153ed0bf339c39533c314e7b1ec1946fd20b0f8324d3b956e8834db'),
'transactionIndex': 3, 'blockHash':
HexBytes('0x60610ff99eaa20ed9c270f166692331f88371d6e7f51e2043ec92725a163406f'),
'logIndex': 2, 'removed': False}), AttributeDict({'address':
'0x8619d80FB0141ba7F184CbF22fd724116D9f7ffC', 'topics':
[HexBytes('0x458f5fa412d0f69b08dd84872b0215675cc67bc1d5b6fd93300a1c3878b86196'),
HexBytes('0x000000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b'),
HexBytes('0x000000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b')],
'data':
HexBytes('0x000000000000000000000000000000000000000000000008ac72670e05a2e800000
00000000000000000000000000000000000000000000000036c56722e80000000000000000000000
00000000000000000000040973577cf6e3dee8b09677'), 'blockNumber': 359934146,
'transactionHash':
HexBytes('0x9d3042698153ed0bf339c39533c314e7b1ec1946fd20b0f8324d3b956e8834db'),
'transactionIndex': 3, 'blockHash':
HexBytes('0x60610ff99eaa20ed9c270f166692331f88371d6e7f51e2043ec92725a163406f'),
'logIndex': 3, 'removed': False}), AttributeDict({'address':
'0x794a61358D6845594F94dc1DB02A252b5b4814aD', 'topics':
[HexBytes('0x804c9b842b2748a22bb64b345453a3de7ca54a6ca45ce00d415894979e22897a'),
HexBytes('0x000000000000000000000000da10009cbd5d07dd0cecc66161fc93d7c9000da1')],
'data':
HexBytes('0x000000000000000000000000000000000000000000023b848e9f0cf46d8a8933000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000034ca0a2835aa0d14ed7a300000000000000000000000000000000000000
000003bf39477c620cc99ff9d42c000000000000000000000000000000000000000040973577cf6e
3dee8b09677'), 'blockNumber': 359934146, 'transactionHash':
HexBytes('0x9d3042698153ed0bf339c39533c314e7b1ec1946fd20b0f8324d3b956e8834db'),
'transactionIndex': 3, 'blockHash':
HexBytes('0x60610ff99eaa20ed9c270f166692331f88371d6e7f51e2043ec92725a163406f'),
'logIndex': 4, 'removed': False}), AttributeDict({'address':
'0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1', 'topics':
[HexBytes('0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'),
HexBytes('0x000000000000000000000000082e64f49ed5ec1bc6e43dad4fc8af9bb3a2312ee'),
HexBytes('0x000000000000000000000000005b823270e3719cde8669e5e5326b455eaa8a350b')],
'data':
HexBytes('0x000000000000000000000000000000000000000000000008ac7230489e80000'),

'blockNumber': 359934146, 'transactionHash':
HexBytes('0x9d3042698153ed0bf339c39533c314e7b1ec1946fd20b0f8324d3b956e8834db'),
'transactionIndex': 3, 'blockHash':
HexBytes('0x60610ff99eaa20ed9c270f166692331f88371d6e7f51e2043ec92725a163406f'),
'logIndex': 5, 'removed': False}), AttributeDict({'address':
'0x794a61358D6845594F94dc1DB02A252b5b4814aD', 'topics':
[HexBytes('0xb3d084820fb1a9decffb176436bd02558d15fac9b0ddfed8c465bc7359d7dce0'),
HexBytes('0x000000000000000000000000da10009cbd5d07dd0cecc66161fc93d7c9000da1'),
HexBytes('0x000000000000000000000000b823270e3719cde8669e5e5326b455eaa8a350b'),
HexBytes('0x0000000000000000000000000000000000000000000000000000000000000000')],
'data':
HexBytes('0x000000000000000000000000b823270e3719cde8669e5e5326b455eaa8a350b0000
00000000000000000000000000000000000000000008ac7230489e800000000000000000000000
00000000000000000000000000000000000000000000200000000000000000000000000000000000
00000034ca0a2835aa0d14ed7a30'), 'blockNumber': 359934146, 'transactionHash':
HexBytes('0x9d3042698153ed0bf339c39533c314e7b1ec1946fd20b0f8324d3b956e8834db'),
'transactionIndex': 3, 'blockHash':
HexBytes('0x60610ff99eaa20ed9c270f166692331f88371d6e7f51e2043ec92725a163406f'),
'logIndex': 6, 'removed': False})], 'logsBloom':
HexBytes('0x0008000000004000000000000000000000000000000000000000000000000000000
00000000000000000000004000100000004000040000000000000000008000000000000000100
00080000000000004000000000000000004000000000000000020000000000000000008000000
00000000000000020100000000000000000000000000001000000000000000000800000000000
00000000000000000000000000000000000000001000010000000000000080000000000001000200
00020000010000000200000008200000000000040000000000000200000000000200000000008
00000001000000000000000000000000000000000000'), 'status': 1, 'timeboosted': False, 'to':
'0x794a61358D6845594F94dc1DB02A252b5b4814aD', 'transactionHash':
HexBytes('0x9d3042698153ed0bf339c39533c314e7b1ec1946fd20b0f8324d3b956e8834db'),
'transactionIndex': 3, 'type': 0})
✅ DAI borrow successful!
✅ Operation 'borrow' recorded. Success rate: 100.0%
Next operation available in 60s

📊 AUTONOMOUS TASK SUMMARY:
Operations completed: 5
Performance score: 0.600
Run: 1, Iteration: 5
Logged performance: Run 1, Iteration 5, Metric: 0.6000000000000001

Analyzing performance for self-improvement...
Average recent performance: 0.4530
Configuration saved.
Logged improvement: Increasing exploration rate due to low performance.

🤝 COLLABORATIVE STRATEGY ANALYSIS:

⚠️ Agent missing get_recent_performance method: 'ArbitrumTestnetAgent' object has no attribute 'get_recent_performance'
📭 No pending proposals
✅ Iteration 5 completed successfully. Waiting for next cycle.
💤 Sleeping for 45 seconds before next iteration...

--- Autonomous Agent Loop: Iteration 6 ---

🚀 STARTING AUTONOMOUS RUN: 1, ITERATION: 6
==========
📋 Step 1: Network and wallet validation...
🌐 Network Congestion Analysis:
  Base Fee: 0.01 gwei
  Current Gas: 0.01 gwei
  Congestion Ratio: 1.00x
🌐 Network Connected: Block 359934326, Gas Price: 0.01 Gwei
🔗 RPC Endpoint: https://arbitrum-one.public.blastapi.io
⛓️ Chain ID: 42161 (Expected: 42161)
✅ Aave Pool accessible: Revision 8
✅ Network validation passed: Connected and validated
💰 Current ETH balance: 0.003760 ETH
📊 Step 4: Checking Aave account status...
  Health Factor: 2.6220
  Collateral: $218.24
  Available Borrows: $91.81
✅ Conditions favorable for DAI borrowing operations
📊 Predicted success rate: 90.0%
⏰ Operation in cooldown. 15s remaining
⏰ Operations in cooldown period

📊 AUTONOMOUS TASK SUMMARY:
  Operations completed: 3
  Performance score: 0.300
  Run: 1, Iteration: 6
Logged performance: Run 1, Iteration 6, Metric: 0.30000000000000004

🤝 COLLABORATIVE STRATEGY ANALYSIS:
⚠️ Agent missing get_recent_performance method: 'ArbitrumTestnetAgent' object has no attribute 'get_recent_performance'
📭 No pending proposals
✅ Iteration 6 completed successfully. Waiting for next cycle.
💤 Sleeping for 45 seconds before next iteration...

--- Autonomous Agent Loop: Iteration 7 ---

🚀 STARTING AUTONOMOUS RUN: 1, ITERATION: 7

==========
📋 Step 1: Network and wallet validation...
🌐 Network Congestion Analysis:
Base Fee: 0.01 gwei
Current Gas: 0.01 gwei
Congestion Ratio: 1.00x
🌐 Network Connected: Block 359934507, Gas Price: 0.01 Gwei
🔗 RPC Endpoint: https://arbitrum-one.public.blastapi.io
⛓️ Chain ID: 42161 (Expected: 42161)
✅ Aave Pool accessible: Revision 8
✅ Network validation passed: Connected and validated
💰 Current ETH balance: 0.003760 ETH
📊 Step 4: Checking Aave account status...
Health Factor: 2.6221
Collateral: $218.25
Available Borrows: $91.82
✅ Conditions favorable for DAI borrowing operations
📊 Predicted success rate: 90.0%
💰 Safe borrow calculation:
Available: $91.82
Safe amount (15%): $10.00
🎯 Attempting safe DAI borrow: $10.00
🏦 Enhanced Borrow Manager: STRICT DAI-ONLY BORROWING
💰 Target amount: $10.00 DAI
🎯 DAI address: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
❌ USDC fallback: DISABLED
✅ Confirmed DAI address: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
💱 USD to DAI conversion: $10.00 = 10000000000000000000 DAI wei
🔍 DAI-specific borrow validation for $10.00
✅ DAI address confirmed: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
🔍 Enhanced validation for $10.00 borrow...
✅ Network connected - Block: 359934508
🔍 Enhanced Validation Results:
Total Collateral: $218.25
Total Debt: $65.11
Available Borrows: $91.82
Health Factor: 2.6221
ETH Balance: 0.003760 ETH
✅ All validation checks passed
🔄 DAI borrow attempt 1/3
🎯 Only borrowing DAI: 0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1
🔄 Borrow attempt 1/3
🔧 Getting optimized gas for aave_borrow in normal conditions
⛽ Network gas conditions:
Current gas price: 0.01 gwei
Base fee: 0.01 gwei

Network congestion factor: 1.00
Total gas multiplier: 1.32
✅ EIP-1559 gas params:
Gas limit: 600,000
Max fee: 0.51 gwei
Priority fee: 0.50 gwei
✅ Borrow successful:
0xb0f9d0f110a7c4c42598349c11bf64ba36333dbad5950be3a6340d786cd28b98
✅ DAI borrow successful on attempt 1
✅ SUCCESS: Borrowed $10.00 DAI
🔗 Transaction hash: AttributeDict({'blockHash':
HexBytes('0x845908d354075edcff65dd4a97b11b6f461e118d537d2331d3f134e3d4b1b553'),
'blockNumber': 359934518, 'contractAddress': None, 'cumulativeGasUsed': 1706618,
'effectiveGasPrice': 10000000, 'from': '0x5B823270e3719CDe8669e5e5326B455EaA8a350b',
'gasUsed': 372703, 'gasUsedForL1': '0x1435', 'l1BlockNumber': '0x15e6d3d', 'logs':
[AttributeDict({'address': '0x8619d80FB0141ba7F184CbF22fd724116D9f7ffC', 'topics':
[HexBytes('0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'),
HexBytes('0x0000000000000000000000000000000000000000000000000000000000000000'),
HexBytes('0x000000000000000000000000000000000000000000000000000000000000000000'),
'data':
HexBytes('0x000000000000000000000000000000000000000000000000000000000000000000'),
'blockNumber': 359934518, 'transactionHash':
HexBytes('0xb0f9d0f110a7c4c42598349c11bf64ba36333dbad5950be3a6340d786cd28b98'),
'transactionIndex': 8, 'blockHash':
HexBytes('0x845908d354075edcff65dd4a97b11b6f461e118d537d2331d3f134e3d4b1b553'),
'logIndex': 10, 'removed': False}), AttributeDict({'address':
'0x8619d80FB0141ba7F184CbF22fd724116D9f7ffC', 'topics':
[HexBytes('0x458f5fa412d0f69b08dd84872b0215675cc67bc1d5b6fd93300a1c3878b86196'),
HexBytes('0x000000000000000000000000000000000000000000000000000000000000000000'),
HexBytes('0x000000000000000000000000000000000000000000000000000000000000000000'),
'data':
HexBytes('0x000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000

000003bf394f7dc712a3ef9cdae1000000000000000000000000000000000000000040973643c116
75146726cda'), 'blockNumber': 359934518, 'transactionHash':
HexBytes('0xb0f9d0f110a7c4c42598349c11bf64ba36333dbad5950be3a6340d786cd28b98'),
'transactionIndex': 8, 'blockHash':
HexBytes('0x845908d354075edcff65dd4a97b11b6f461e118d537d2331d3f134e3d4b1b553'),
'logIndex': 12, 'removed': False}), AttributeDict({'address':
'0xDA10009cBd5D07dd0CeCc66161FC93D7c9000da1', 'topics':
[HexBytes('0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef'),
HexBytes('0x000000000000000000000000082e64f49ed5ec1bc6e43dad4fc8af9bb3a2312ee'),
HexBytes('0x000000000000000000000000b823270e3719cde8669e5e5326b455eaa8a350b')],
'data':
HexBytes('0x0000000000000000000000000000000000000000000000008ac7230489e80000'),
'blockNumber': 359934518, 'transactionHash':
HexBytes('0xb0f9d0f110a7c4c42598349c11bf64ba36333dbad5950be3a6340d786cd28b98'),
'transactionIndex': 8, 'blockHash':
HexBytes('0x845908d354075edcff65dd4a97b11b6f461e118d537d2331d3f134e3d4b1b553'),
'logIndex': 13, 'removed': False}), AttributeDict({'address':
'0x794a61358D6845594F94dc1DB02A252b5b4814aD', 'topics':
[HexBytes('0xb3d084820fb1a9decffb176436bd02558d15fac9b0ddfed8c465bc7359d7dce0'),
HexBytes('0x000000000000000000000000da10009cbd5d07dd0cecc66161fc93d7c9000da1'),
HexBytes('0x000000000000000000000000b823270e3719cde8669e5e5326b455eaa8a350b'),
HexBytes('0x0000000000000000000000000000000000000000000000000000000000000000')],
'data':
HexBytes('0x000000000000000000000000b823270e3719cde8669e5e5326b455eaa8a350b0000
0000000000000000000000000000000000000000000008ac7230489e800000000000000000000000
0000000000000000000000000000000000000000000020000000000000000000000000000000000
00000034cb48f9e404c9dfdbfaa4'), 'blockNumber': 359934518, 'transactionHash':
HexBytes('0xb0f9d0f110a7c4c42598349c11bf64ba36333dbad5950be3a6340d786cd28b98'),
'transactionIndex': 8, 'blockHash':
HexBytes('0x845908d354075edcff65dd4a97b11b6f461e118d537d2331d3f134e3d4b1b553'),
'logIndex': 14, 'removed': False})], 'logsBloom':
HexBytes('0x0008000000004000000000000000000000000000000000000000000000000000
0000000000000000000000004000100000000400004000000000000000000800000000000000100
0008000000000000400000000000000000400000000000000002000000000000000000008000000
0000000000000000201000000000000000000000000000001000000000000000000800000000000
0000000000000000000000000000000000000000001000010000000000000008000000000001000200
0002000001000000002000000008200000000000400000000000000200000000000200000000008
0000000100000000000000000000000000000000000000'), 'status': 1, 'timeboosted': False, 'to':
'0x794a61358D6845594F94dc1DB02A252b5b4814aD', 'transactionHash':
HexBytes('0xb0f9d0f110a7c4c42598349c11bf64ba36333dbad5950be3a6340d786cd28b98'),
'transactionIndex': 8, 'type': 0})
✅ DAI borrow successful!
✅ Operation 'borrow' recorded. Success rate: 100.0%
Next operation available in 60s

📊 AUTONOMOUS TASK SUMMARY:

Operations completed: 5
Performance score: 0.600
Run: 1, Iteration: 7
Logged performance: Run 1, Iteration 7, Metric: 0.6000000000000001

🤝 COLLABORATIVE STRATEGY ANALYSIS:
⚠️ Agent missing get_recent_performance method: 'ArbitrumTestnetAgent' object has no attribute 'get_recent_performance'
📭 No pending proposals
✅ Iteration 7 completed successfully. Waiting for next cycle.
💤 Sleeping for 45 seconds before next iteration...